

Nicholas Eisenberg, Ph.D.

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Experience

Senior Scientist - Machine Learning

June 2022 - Present

Nevada National Security Site

- **LLM Inference Server (Piccata):** Architected, deployed, and administered a multi-user LLM platform on a RHEL GPU server, exposing internally hosted models through OpenAI-compatible vLLM APIs. Built a FastAPI gateway for authentication, routing, usage tracking, and persistence. Developed `libserverctl`, a python library for user, API-key, certificate, and usage administration, and built a `Streamlit` portal for access requests and provisioning API tokens and client certificates. Deployed the platform with `Podman`, `Nginx` reverse proxying and TLS/mTLS.
- **LLM-Assisted Image Annotation and Retrieval (Photo-Pilot):** Designed and deployed a `Streamlit` application that enables users to annotate and tag images with the assistance of multimodal LLMs hosted on Piccata. The annotations and metadata were persisted in a SQL database, and the portal allowed the users to query the resulting image collection. Deployed the application on a RHEL server using `Podman`, with `Nginx` providing reverse proxying.
- **Python LLM Client Library (ChatLib):** Developed and maintained an internal Python library built on the OpenAI SDK that simplified interaction with internally hosted LLMs, providing conversation management, streaming, multimodal text and image inputs, response formatting, and a consistent interface for scientific and engineering applications.
- **Internal Scientific Software Monorepo:** Maintained a large internal monorepo which supported ML and scientific applications, including shared Python libraries, dependency and release workflows, automated testing, CI/CD, and Linux deployment infrastructure.
- **Retrieval-Augmented Generation (RAG):** Designed and deployed a RAG system for internal Python/C++ codebases and technical documents, implementing language-aware ingestion, semantic embeddings, HNSW vector retrieval, and locally hosted LLM inference.

Technical Skills

Programming Languages: Python, Bash, Lua, SQL, Cypher, $\LaTeX$

Libraries: PyTorch, Pandas, FastAPI, SQLAlchemy, Streamlit, Matplotlib, vLLM

Tools / Infrastructure: Linux (RHEL, Ubuntu), Podman, Docker, Nginx, Git, CI/CD, SLURM, AWS

Open-Source Projects

iron.nvim - Sole Maintainer

May 2024 - Present

github.com/Vigemus/iron.nvim

Sole maintainer of a cross-platform Neovim plugin written in Lua for interactive REPL integration. I owned the project lifecycle including feature development, code review, releases, testing, issue triage, and technical support across Linux, macOS, and Windows.

Education

Auburn University - Ph.D.

June 2022

Research Area: Stochastic Partial Differential Equations

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. *Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics*. Stoch PDE: Anal Comp (2022).

- 2: Eisenberg, N., Chen, L. *Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term.*
J Theor Probab (2024).